# MSG No. 86.68 $P4_2/n1'$ [ Type II, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `-4..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,4,6,7,9,12,14,15]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[2,3,5,8,10,11,13,16]` |

Table 2: Wyckoff site: `2b`, site symmetry: `-4..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1,4,6,7,9,12,14,15]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,5,8,10,11,13,16]` |

Table 3: Wyckoff site: `4c`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,5,9,13]` |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,6,10,14]` |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[3,7,11,15]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[4,8,12,16]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,5,9,13]` |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | `[2,6,10,14]` |
| 3 | $\left[\frac{1}{2}, 0, 0\right]$ | `[3,7,11,15]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[4,8,12,16]` |

Table 5: Wyckoff site: `4e`, site symmetry: `2..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1,4,9,12]` |
| 2 | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[2,3,10,11]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[5,8,13,16]` |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[6,7,14,15]` |

Table 6: Wyckoff site: `4f`, site symmetry: `2..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,4,9,12]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[2,3,10,11]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | `[5,8,13,16]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[6,7,14,15]` |

Table 7: Wyckoff site: `8g`, site symmetry: `11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1,9]` |
| 2 | $\left[-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$ | `[2,10]` |
| 3 | $\left[y+\frac{1}{2}, -x, z+\frac{1}{2}\right]$ | `[3,11]` |
| 4 | $\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$ | `[4,12]` |
| 5 | $[-x, -y, -z]$ | `[5,13]` |
| 6 | $\left[y, \frac{1}{2}-x, \frac{1}{2}-z\right]$ | `[6,14]` |
| 7 | $\left[\frac{1}{2}-y, x, \frac{1}{2}-z\right]$ | `[7,15]` |
| 8 | $\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$ | `[8,16]` |